

Buying Certainty for Convex Region Libraries: an Experimental Record

strict regeneration, SOS certification, cross-machine verification,
and why the optimizer amplifies volume error

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Abstract

This record documents four experiments run after the region-soundness measurement: (A) cross-machine verification of the reference-answer audit; (B) full min-clearance depth profiles of all 18 colliding task-pair references; (C) strict regeneration of the 219-region bimanual library with the Bernoulli gate enforced at generation (~ 40 node-hours); and (D) a C-IRIS (SOS-certified) library composed with our interface-sampling layer on a 2-DoF testbed. Two laws organize the results. *The amplification law*: a 3.2% volume error becomes a 12.5% answer error on random queries and a 62% answer error on task queries, because the optimizer is an argmin over the modeled set—an active search for exactly the model surplus that shortens paths—not a volume-uniform sampler. *The connectivity law*: all three routes to volume-level certainty that we tested (a stricter sampling gate, SOS certificates, uniform margins) destroy the connectivity that planning needs before they deliver soundness where tasks live; the strict gate accepts 47/222 regions that are pairwise disjoint (0 portals), refuting our own registered prediction in the strongest direction. Every number below carries its generating artifact.

1 Scope and headline numbers

All experiments target the scaled bimanual scene ($2 \times$ KUKA iiwa14 + WSG grippers + shelves, 14 DoF, 903 filtered collision pairs) and its 219-region IrisZo library, except campaign D (2-DoF planar arm, the regime where SOS certification is practical). Cluster jobs ran on CIT-TUM-HN (Slurm; nodes 128–256 cores); MOSEK-dependent SOS work ran locally (license binding).

campaign	jobs	wall	artifact
A: cross-machine e9 audit	425773	~ 1 h	e9_verify_gcsstar_cluster.json
B: depth profiles	425802	< 1 h	e9b_depth_profiles.json
C: strict regeneration	425771/978, 426330(+merge)	~ 40 node-h	e10_regen219_report.json
D: C-IRIS composition	local (MOSEK)	~ 10 min	e15_ciris.json, e15b_*.json

2 The amplification law

Three percentages measure three different things:

quantity	value	measure
interface contamination	3.2% (64/2000)	volume-uniform sampling
random-query reference optima colliding	12.5% (2/16)	argmin, generic endpoints
task-query reference optima colliding	62% (18/29)	argmin, task endpoints

If planner answers were volume-uniform darts, 3.2% dirt would produce $\approx 3\text{--}5\%$ dirty answers. The measured factor-20 amplification decomposes into three mechanisms:

(i) Taut paths patrol the boundary shell. A shortest path hugs obstacle boundaries and cuts corners; the residual pockets live exactly in the boundary shell where the generator’s cuts stopped. A volume-uniform sample visits the shell occasionally; a taut path rides it for most of its length.

(ii) The argmin is an active search for model surplus.

Proposition 1 (optimizers exploit surplus). *Let P be the modeled free polytope union, $P \cap \mathcal{F}$ the true one, and let $m(\cdot)$ denote the optimal path cost through a set. If $m(P) < m(P \cap \mathcal{F})$ —the through-pocket shortcut is genuinely shorter—then every optimum of the modeled problem is infeasible, and the more exact the solver, the more certainly it is found.*

The solver is not unlucky; it minimizes over the model, so wherever the model errs optimistically and the error shortens paths, the argmin lands there with probability one. A navigation analogy: if the map contains one nonexistent tunnel that saves two minutes, the shortest-path router takes it 100% of the time, regardless of how small the tunnel is or how accurate the rest of the map may be.

(iii) Task queries live in the dirtiest real estate. Grasp configurations hug the shelves—the same cramped geometry where the generator’s iteration caps truncated and left the deepest pockets—and task corridors ride the grasp-seeded regions. The query distribution concentrates paths in the contaminated shell. The chain: volume 3.2% $\xrightarrow{\times 4 \text{ (i)+(ii)}}$ random answers 12.5% $\xrightarrow{\times 5 \text{ (iii)}}$ task answers 62%.

Measure mismatch, stated once. Library audits sample the *uniform* measure; the planner samples the *argmin* measure, which is supported precisely on cost-reducing model errors. The two measures are misaligned by construction, so no amount of uniform auditing bounds the argmin failure rate. Campaign D provides a miniature: a 2-DoF library passes 3,987 uniform audit samples with zero failures, yet 1 of 30 optima collides (Sec. 6). The only test aligned with the argmin measure is a test of the argmin itself—validating the returned path, which is what the runtime certificate layer does.

3 Campaign A: the audit is machine-independent

The 16-query reference audit (bit-exact cost reproduction $\rightarrow$ chain-SOCP argmin $\rightarrow$ dual-mechanism validation $\rightarrow$ pair classification) was re-run unmodified on the cluster.

- All 16 verdicts reproduce. The two confirmed violations reproduce with *identical colliding-probe counts*: q8 286/2000 (gripper–shelf, -63.5 mm) and q13 581/2000 (arm–arm graze, -0.3 mm) on both machines.

- q5’s rerun deviation (5.1×10^{-3}) is identical on both machines (deterministic solver noise), and stays CLEAN.
- q14 resolves to *unstable-and-colliding*: the local rerun returns no path; the cluster rerun returns cost 10.2139 against the stored 10.0395 (1.7% drift, far above solver noise), and that variant itself collides (128/2000 probes, gripper–grripper -1.6 mm). Under our bit-exact standard it is *not* counted; the confirmed tally stays 2/16. The variant’s collision is consistent with Sec. 2(ii): whatever optimum the solver lands on hugs the contaminated boundary.

4 Campaign B: depth profiles of the 18 colliding task pairs

Recipe: reproduce each reference with the sweep’s exact configuration (`gcs_star(timeout=60, K=1, max_revisits=1, seed=0)`), re-derive all waypoints via the portal-chain SOCP whose optimum *is* the reported cost, then walk the path with a 4,000-point min-clearance scan. A recorded pitfall: an earlier attempt used the full-graph relaxation solver instead (3h without finishing one pair); the reference recipe above reproduces every stored cost to the last digit in 0–8s per pair (two pairs at $\sim 10^{-16}$, floating noise).

from	to	cost	worst (mm)	at (%)	dirty (%)	ends (mm)
top_shelf/top_shelf	bin_R/shelf_1	4.358	−40.4	58	22.9	+38/ +82
top_shelf/top_shelf	bin_R/bin_L	4.809	−41.0	21	10.6	+38/ +100
top_shelf/top_shelf	neutral/neutral	2.039	−7.9	50	7.3	+38/ +100
top_shelf/top_shelf	neutral/shelf_1	2.480	−41.2	60	51.1	+38/ +82
top_shelf/top_shelf	shelf_1/neutral	2.214	−39.7	52	21.8	+38/ +82
top_shelf/bin_L	bin_R/shelf_1	4.864	−30.8	85	12.8	+64/ +82
top_shelf/bin_L	bin_R/bin_L	5.497	−24.3	16	6.5	+64/ +100
top_shelf/bin_L	neutral/shelf_1	3.426	−24.6	71	16.6	+64/ +82
top_shelf/bin_L	shelf_1/neutral	3.085	−39.1	61	28.7	+64/ +82
bin_R/top_shelf	bin_R/shelf_1	1.809	−44.6	68	19.4	+64/ +82
bin_R/top_shelf	neutral/shelf_1	3.985	−48.9	63	27.7	+64/ +82
bin_R/top_shelf	shelf_1/neutral	4.113	−34.1	86	12.9	+64/ +82
bin_R/shelf_1	bin_R/bin_L	4.791	− 89.6	16	14.2	+82/ +100
bin_R/shelf_1	neutral/neutral	4.121	−1.7	60	3.0	+82/ +100
bin_R/shelf_1	shelf_1/neutral	4.445	−33.6	86	13.1	+82/ +82
bin_R/bin_L	neutral/shelf_1	4.787	− 80.3	77	13.8	+100/ +82
bin_R/bin_L	shelf_1/neutral	4.750	− 91.3	86	13.8	+100/ +82
neutral/shelf_1	shelf_1/neutral	3.449	−16.7	60	16.5	+82/ +82

Summary: median worst penetration -39.4 mm, maximum -91.3 mm, three pairs beyond -80 mm. Every violation is a *single contiguous mid-path interval* (16–86% arc position) while both endpoints clear by $+38$ to $+100$ mm: the named grasp configurations are physically valid; the *path* is what enters the shelves. This closes the “query itself invalid” escape (q8’s sampled start had that defect; no task pair does), and retires the “grazing” reading of the first-crossing depths (0–1 mm) recorded by the sweep.

5 Campaign C: strict regeneration collapses the graph

Recipe. Same 222 seeds as the original library (22 grasp + 200 random, same rng), with every knob tightened and the audit enforced as a gate:

	original (shipped defaults)	strict
ε	1%	0.1%
δ	5%	1%
particles	1,000	4,000
outer iterations	3 (cap-truncated admitted)	10
sep.-plane iterations	default	40
mixing steps	default	80
acceptance	none (cap $\Rightarrow$ silent admit)	Bernoulli gate $M = 4,605$, 2 regrowth retries, else drop

Engineering (protocol-neutral). Per-seed atomic checkpoints; 8-way stride sharding (Slurm array); audit/jitter rngs keyed by seed name (outcomes independent of sharding); **parallelism** pinned to the cpu allocation (the default spawns one thread per hardware core and oversubscribes). The campaign crossed two 16 h walls; checkpoint resume made both routine (130 and 217 of 222 preserved, zero recomputation), three tail rounds total. Open-space random seeds are the expensive ones under strict settings (45–100 min per seed; accepted grows median 129 s, p90 454 s); shelf-adjacent seeds fail fast in small regions.

Acceptance.

seed class	accepted	dropped	rate
grasp (shelf-adjacent)	3	19	13.6%
random	44	156	22.0%
total	47	175	21.2%

Of the 47 accepted: 26 passed on the first grow, 15 on the second, 6 on the third—45% owe their acceptance to regrowth retries (survivor selection in action). The surviving grasp regions are `shelf_1/top_shelf`, `shelf_1/shelf_1`, `bin_R/top_shelf`. Accepted regions carry a median of 207 faces (range 89–428).

The gate’s refusals are near-misses. All 559 audit failures across the campaign were killed by a *single* bad sample. The clean streak before that sample has median 74 (consistent with $\sim 1.4\%$ true contamination), p90 1,626, maximum 4,557—one region came within 48 samples of the 4,605 bar. The strict target (0.1%) is simply not what this generator produces in this scene, anywhere: not at the shelves, not in the open.

The zero. The merge step measured the accepted library’s topology:

	original library	strict library
regions	219	47
portals (overlaps)	2,757	0
median Chebyshev radius	0.476	0.240
interface contamination	3.2%	undefined (no interfaces)

The zero was verified independently: all $\binom{47}{2} = 1,081$ region pairs were re-tested by LP (Chebyshev center of the stacked constraint systems); none intersect. The 47 survivors are pairwise disjoint

islands: *the strict library supports no planning at all*—no task pair, no random pair, nothing that crosses a region boundary.

The registered prediction, refuted. We had pre-registered: “on the strictly regenerated library, most of the 13 task pairs our layer honestly reports unsolvable should become solvable.” The outcome is the strongest possible reversal: rather than making task regions clean, the strict contract refuses 86% of them and shrinks the rest until the graph has no edges. Tighter generation does not rescue the tasks; it abolishes the planner.

6 Campaign D: C-IRIS composed with the sampling layer

2-DoF planar arm (2-link URDF, 4 box obstacles, 6 collision pairs), the regime where SOS certification is practical in seconds. Same 7 seeds for both libraries; the same layer on top (`build_topology` → interface classification → area sampling audit → GCS answer → dense ground-truth validation). C-IRIS = `CspaceFreePolytope` binary search + 12 rounds of bilinear alternation (MOSEK), polytopes in the rational $s = \tan((q - q^*)/2)$ space.

	IrisZo (defaults)	C-IRIS (SOS)
generation, 7 regions	5 s	78–91 s (18×)
interface audit	0/1992	0/2000 (<i>certificate honored</i>)
interior audit	0/1995	0/1995
free-space coverage	83.1%	33.3%
overlap edges	12 (connected)	1
30 random queries	21 clean + 1 colliding + 8 no-path	0 solved
+12 seeds (2.7× budget)	—	coverage 61%, 12 portals, 7 components , 0/30

Two findings. First, the deterministic certificate is real and our audit machinery composes with it unchanged—0 dirt on 3,995 samples. Second, connectivity, not certification, is the binding constraint: even at 2.7× the seed budget the certified regions fragment into 7 islands (sizes 6, 6, 3, 1, 1, 1, 1), because the tangent parametrization stretches the outer joint range ($|s| \approx 14$ at $q = \pm 3$) and SOS-conservative boundaries stop neighboring regions at the same walls instead of overlapping. Meanwhile the IrisZo side donated the miniature of Sec. 2: 3,987 uniform audit samples all clean, yet 1 of 30 optima collides—a pocket thinner than the audit resolution, found by the argmin.

Four reproduction pitfalls, recorded: (i) `BinarySearchOptions.scale_max` defaults to 1.0 and pins the region to the seed box; (ii) the binary-search result sits on the certification boundary—back off 2% toward the center or the alternation returns an *empty* history (silently); (iii) the certified set is $\{s \mid Cs \leq d, s_{lo} \leq s \leq s_{up}\}$: the joint-limit rows must ship with the exported polytope (after alternation reshapes C , the bare (C, d) can be unbounded); (iv) “first solve failed” from the GCS layer means the region graph is disconnected—check components before blaming the solver.

7 Synthesis: the connectivity law

Three independent mechanisms for volume-level certainty, three independent collapses of the planning graph:

route	certainty delivered	graph cost (measured)
strict sampling gate ($\varepsilon=0.1\%$, enforced)	probabilistic, $10\times$ tighter	$219 \rightarrow 47$ regions, $2757 \rightarrow 0$ portals
SOS certificates (C-IRIS, 2-DoF)	deterministic	7 islands at $2.7\times$ seeds, 0/30 solved
uniform margins (erosion $\delta=0.05$ rad)	none (contamination $3.4 \rightarrow 2.6\%$)	queries disconnect first
runtime path certificate	deterministic, <i>per answer</i>	none: the graph is untouched

The pattern has one cause: certainty about a d -dimensional volume must be paid wherever the volume touches obstacles, and in cluttered scenes that price is highest exactly where regions must meet (interfaces near shelves). Certainty about a 1-dimensional path is paid only along the path. The scissors close in opposite directions with dimension: contamination grows (0% at 2-DoF, 0.3% at 7, 3.2% at 14) while volume-level certification becomes unaffordable ($18\times$ at 2-DoF already; out of reported practice at 14). The default library is connected but lies (62% of task optima infeasible, to -91 mm); the strict library is honest but mute (0 portals). The only mechanism in this study that keeps the graph *and* returns only verified answers is the runtime certificate layer, whose price is the path’s own length: $\sim 10^2$ clearance calls at 2.3 ms, 0.02–0.06 s per certificate.

8 Reproducibility

item	script	artifact
cross-machine audit	e9_verify_gcsstar.py	e9_verify_gcsstar{,_cluster}.json
depth profiles	e9b_depth_profiles.py	e9b_depth_profiles.json, ref paths npz
strict regeneration	e10_regen219.py (-stride/-merge)	e10_regen219_report.json, bimanual_regions_sc
slurm	slurm/e10_regen_{array,merge}.sbatch	jobs 425771/425978/426330 + 426331
C-IRIS composition	e15_ciris_2dof.py, e15b_ciris_dense.py	e15_ciris.json, e15b_ciris_dense.json

Environment: Drake 1.54 (IrisZo, CspaceFreePolytope), Clarabel for chain SOCPs, MOSEK for SOS and GCS relaxations (local license), Python 3.12. Cluster: CIT-TUM-HN, Slurm, 32 cpus/task for generation shards. All rngs are fixed-seed; strict-regeneration outcomes are independent of sharding by construction (per-seed keyed streams).